

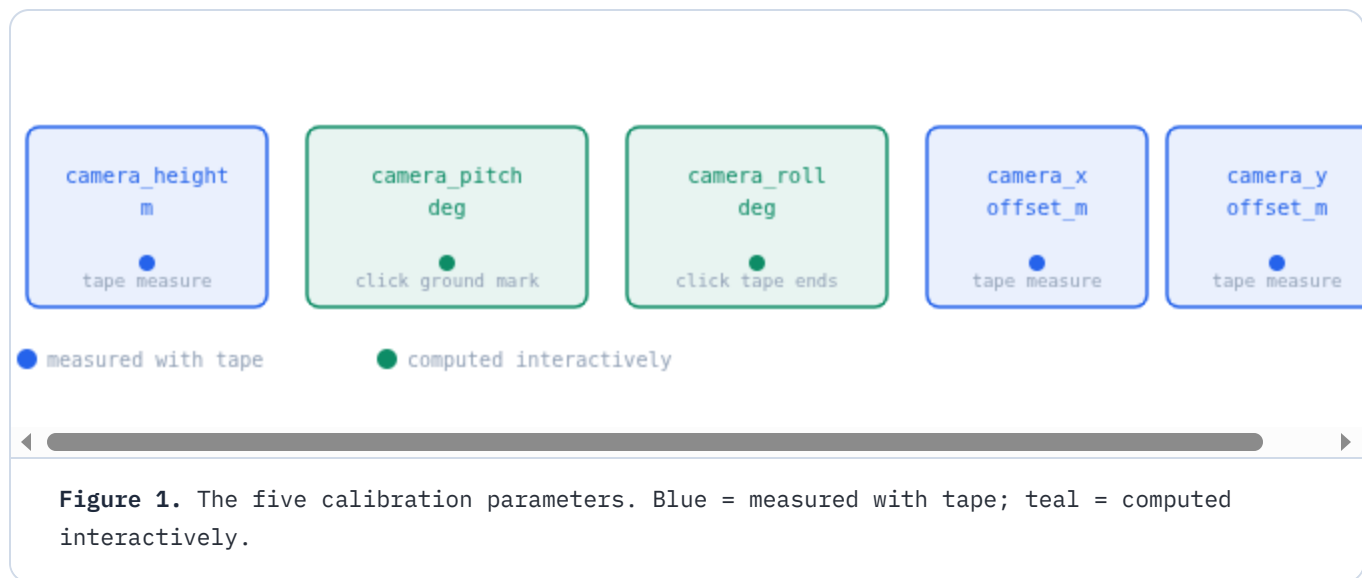
FLAT_WORLD_CAMERA_CALIBRATION_TOOL

Flat-World Camera Calibration

How to determine camera pitch, roll, height, and position offsets for unprojecting image pixels onto a flat ground plane.

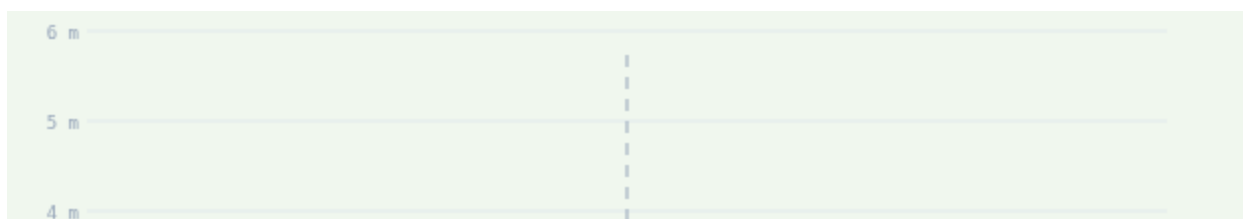
1. What You Are Calibrating

To convert a pixel in the camera image to a point on the ground, you need five parameters. Three are measured physically with a tape measure; two are computed interactively using the tool.



2. Physical Setup

Park the vehicle on flat, level ground. You need two pieces of equipment placed on the ground in front of the camera.



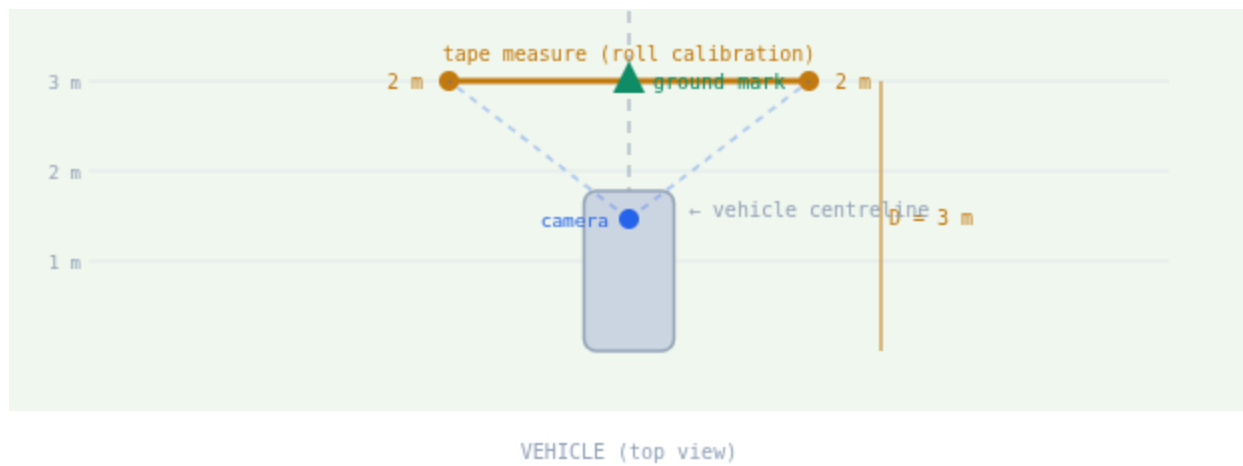


Figure 2. Top-view of the physical setup. The ground mark sits on the vehicle centreline at distance D . The tape measure is perpendicular to the centreline at the same distance D .

- 1 **Measure camera height.** With a tape measure, find the vertical distance from the ground to the camera's optical centre (front of lens). Record this as `camera_height_m` in `calibration_config.yaml`.
- 2 **Choose a mark distance D .** Pick a distance that falls within the camera's field of view. 3–5 m typically works well. Record this as `mark_distance_m`.
- 3 **Place the ground mark.** Set a clearly visible object (cone, tape X, painted dot) on the ground at distance D *directly ahead* of the camera, on the vehicle centreline. Use a tape measure from directly below the camera to the mark.
- 4 **(Optional) Place the roll tape.** Stretch a tape measure on the ground, perpendicular to the vehicle, centred on the centreline, at the same distance D . The tape should be at least 1–2 m wide and clearly visible. Record its width as `tape_width_m`.
- 5 **Capture a calibration image.** With everything in place, save a single frame from the camera to disk (e.g. using `ros2 run image_view image_saver` or any image-capture tool). This is the image you will load into the calibration tool.

Important: the vehicle must be stationary and on flat, level ground while you capture the image. Any pitch of the vehicle body relative to the ground will appear as additional camera pitch and must be accounted for — ideally calibrate on the surface the vehicle typically operates on.

4. Pitch — The Geometry

The side view shows how clicking the ground mark determines the camera pitch angle θ .

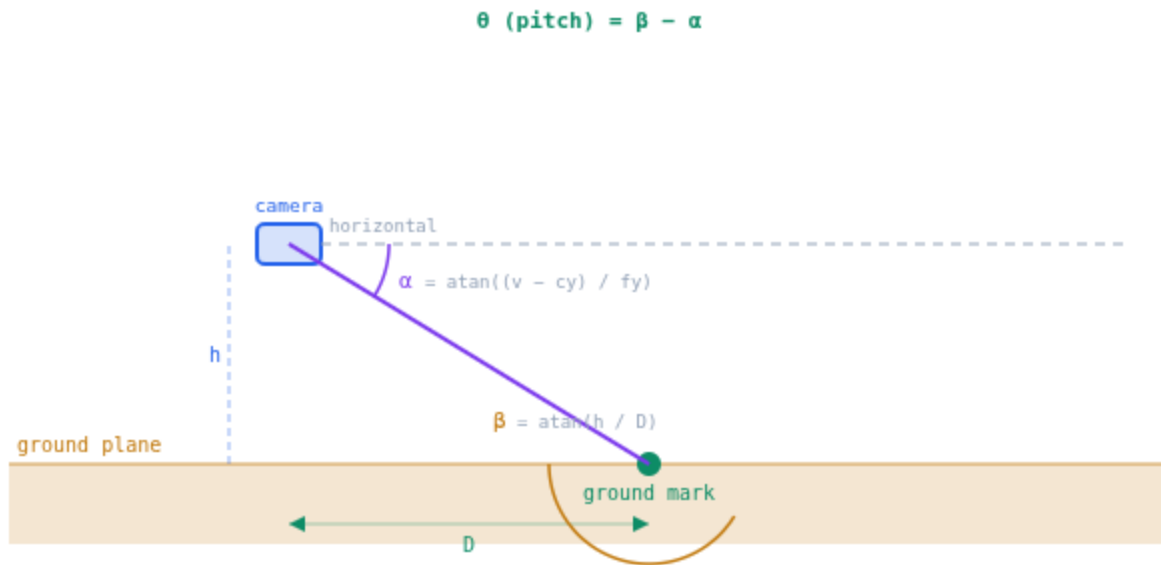


Figure 3. Side-view geometry. The two angles combine to give the camera pitch: the ray angle below the optical axis (from the clicked pixel) and the elevation angle from horizontal to the mark.

The math behind the computation:

```

 $\alpha = \text{atan}((v_{\text{click}} - cy) / fy)$   ← angle of clicked ray
below optical axis
 $\beta = \text{atan}(h / D)$   ← elevation angle to mark below
horizontal
 $\theta = \beta - \alpha$   ← camera pitch (positive = tilted downward)

```

If the mark appears *above* the image centre ($v_{\text{click}} < cy$), α is negative and the computed pitch is larger — this means the camera is tilted down more than the angle to the mark alone would suggest.

4. Roll — The Geometry

Without roll, two points at the same ground distance D (but different lateral positions) project to the same image row v . Any roll tilts this horizontal alignment, producing a slope in the image.



Figure 4. With zero roll (left), the tape ends project to the same image row. With roll (right), the tape endpoints have different v values – the slope gives the roll angle.

```
 $\varphi = \text{atan2}(v_{\text{right}} - v_{\text{left}}, u_{\text{right}} - u_{\text{left}})$  ← camera  
roll (positive = clockwise, right side down)
```

5. Using the Tool

- 1 **Edit the config.** Fill in `calibration_config.yaml` with the image path, intrinsics path, measured height, mark distance, tape width, and vehicle offsets.
- 2 **Launch the tool.**

```
cd flat_world_camera_calibration_tool
python flat_world_calibrate.py
```
- 3 **Calibrate pitch.** Press `p` to enter pitch mode. Click precisely on the centre of the ground mark in the image. The computed pitch appears in the status bar immediately.
- 4 **Calibrate roll (optional).** Press `r`. Click the left end of the tape, then the right end. Roll is computed automatically.
- 5 **Verify with the grid.** Press `g` to show a projected ground grid. Each grid point shows where the corresponding real-world distance and lateral position *should* appear in the image. The highlighted dot at (D, 0) should land exactly on your ground

mark.

- 6 **Save.** Press `s` . The tool writes `flat_world_extrinsic.yaml` with all five parameters.

6. Output File

`flat_world_extrinsic.yaml` contains everything needed by the road centerline node to unproject image pixels onto the ground plane:

```
camera_height_m: 1.2000 # tape-measured
camera_pitch_deg: 15.3420 # computed from pitch click
camera_roll_deg: 0.2100 # computed from tape clicks
camera_x_offset_m: 0.5000 # entered in config
camera_y_offset_m: 0.0000 # entered in config
```

7. Keyboard Reference

KEY	ACTION
p	Pitch mode — next click sets the ground mark
r	Roll mode — click left tape end, then right tape end
g	Toggle ground verification grid overlay
u	Toggle lens undistortion (requires distortion coefficients in intrinsics)
s	Save calibration to flat_world_extrinsic.yaml
q / Esc	Quit

Figure 5. All keyboard controls.